

Software Development – Files

Python programming exercises

1. Write a Python program that will open a file named `this.txt` and write every other line into the file `that.txt`
2. Create a file `words.txt` that contains a paragraph of random words (approximately 100 words long). Next, create a program that prompts for the file name then iterates through each word in this file and counts the frequency of each letter (a through z) and stores the values in a dictionary. Make each letter lowercased and ignore punctuation.
3. Create a test file with a single sentence of 20 words. Read the file, then insert carriage-return characters (`\n`) and write the test to a new text file that will be composed of four lines of five words.
4. Write a program that opens two files: one for reading, the other for writing. The program reads, one line at a time, from the file named `input.txt`. The line is stripped (to remove the carriage return), and then iterated through, one character at a time. We add each character to the left of a new string called `new str`, thus reversing the line. After processing the line, we write `new str` to a file called `output.txt`
5. Write a program that prompts for three numbers. Divide the first number by the second number and add that result to the third number. Using exceptions check for the following errors: `ValueError`, and `ZeroDivisionError`.
6. Write a program using a try-except construct, in the context of user interaction. The goal of the program is to read a line, indicated by a line number, from a file. The user is required to provide both the file name and the line number.